

BALLARAT, AustraliaWMO: **948520**Lat: **37.5128S**Lon: **143.7911E**Elev: **1429**StdP: **13.95**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.7	33.8	28.6	23.8	51.1	30.8	26.4	46.8	32.6	46.2	29.8	47.5	3.1	50	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	91.7	63.6	87.2	62.6	83.4	61.7	68.1	79.5	66.0	78.5	64.3	77.4	16.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	98.2	70.0	62.6	89.4	68.5	60.2	82.2	67.0	33.2	78.6	31.5	78.3	30.2	77.5	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	24.9	22.5	27.1	100.5	1.8	4.0	25.8	103.4	24.7	105.7	23.7	108.0	22.3	110.9		
(5)			26.8	71.8	1.9	4.5	25.4	75.0	24.3	77.7	23.2	80.2	21.8	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	65.3	65.4	61.1	55.0	49.2	45.3	44.1	45.1	48.4	52.4	57.3	61.1		
	DBStd	9.59	7.90	7.41	7.11	5.76	4.41	3.42	3.22	3.86	4.75	6.15	7.23	7.66		
	HDD50	715	0	0	2	17	68	144	184	160	85	41	11	3		
	HDD65	4302	93	80	164	304	489	590	648	617	498	395	252	172		
	CDD50	2208	476	430	348	167	44	4	1	8	37	115	231	347		
	CDD65	320	104	90	45	4	0	0	0	0	0	3	22	52		
	CDH74	4266	1415	1057	534	44	0	0	0	0	1	62	317	836		
	CDH80	1671	639	422	171	6	0	0	0	0	0	10	84	339		
(14) Wind	WSAvg	11.7	12.8	12.6	11.8	10.5	10.6	11.0	12.1	11.9	11.9	11.9	11.7	12.0		
(15) Precipitation	PrecAvg	26.8	1.7	1.5	1.5	2.1	2.5	2.4	2.6	2.9	2.8	2.7	2.1	1.9		
	PrecMax	39.2	7.4	9.4	5.2	5.4	6.8	6.7	4.8	5.4	5.9	7.6	5.2	5.4		
	PrecMin	13.5	0.2	0.0	0.2	0.4	0.4	0.6	0.8	0.5	0.7	0.5	0.7	0.1		
	PrecStd	5.4	1.3	1.7	1.1	1.3	1.4	1.2	1.0	1.0	1.2	1.4	1.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.6	95.9	91.3	79.0	67.0	57.6	55.9	61.9	70.0	80.6	86.9	95.0		
		MCWB	65.1	65.3	62.9	58.1	53.9	51.2	50.2	53.0	56.3	57.9	61.6	62.7		
	2%	DB	92.5	89.4	84.4	73.8	62.4	54.9	53.0	57.7	64.6	74.3	82.1	88.4		
		MCWB	64.2	64.1	61.4	56.8	53.7	50.2	48.5	50.3	54.3	57.2	60.5	62.1		
	5%	DB	86.8	85.4	79.9	70.1	59.4	53.1	51.3	54.7	60.6	69.3	77.3	83.1		
		MCWB	62.8	62.8	60.6	55.6	52.6	49.3	47.8	48.6	52.2	55.8	59.6	60.7		
	10%	DB	82.0	81.0	75.2	66.5	57.0	51.3	49.7	52.1	57.6	64.8	72.1	77.7		
		MCWB	61.6	62.2	59.7	54.8	51.9	48.6	46.8	47.4	50.9	54.2	58.2	59.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.5	71.0	67.1	61.6	58.1	54.0	52.5	54.1	58.4	62.5	67.0	68.3		
		MCDB	77.7	83.0	77.6	71.5	60.6	55.2	53.6	59.9	66.4	71.9	77.3	77.4		
	2%	WB	68.2	68.0	64.6	59.3	55.8	51.6	50.1	51.3	55.6	59.5	63.9	65.1		
		MCDB	80.2	79.6	75.7	67.1	59.4	53.3	51.9	55.6	62.1	69.0	74.2	77.5		
	5%	WB	65.8	65.8	62.7	57.7	54.0	50.2	48.5	49.6	53.4	57.3	61.5	63.0		
		MCDB	81.0	77.6	73.7	65.7	58.0	52.0	50.1	53.2	59.5	66.4	72.2	77.9		
	10%	WB	63.7	64.0	60.9	56.2	52.4	49.0	47.4	48.1	51.4	55.3	59.4	61.1		
		MCDB	78.6	76.8	72.6	64.1	56.0	50.8	49.0	51.4	56.6	63.3	69.9	74.6		
(35) Mean Daily Temperature Range	MDBR		24.4	22.9	20.5	18.2	13.6	11.3	10.9	13.2	15.8	18.9	20.5	23.8		
	5% DB	MCDBR	33.1	30.7	28.3	25.2	17.6	14.4	13.1	16.8	21.3	27.4	30.2	33.8		
		MCWBR	10.8	10.5	10.0	11.1	10.4	10.1	9.2	10.5	12.5	13.8	12.3	12.4		
	5% WB	MCDBR	26.3	24.2	22.8	19.9	14.2	11.3	11.1	15.1	18.6	22.9	24.1	26.4		
		MCWBR	11.3	10.7	10.0	10.4	9.4	8.8	8.5	10.1	11.8	13.1	11.9	12.0		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.328	0.311	0.302	0.294	0.284	0.283	0.292	0.311	0.321	0.329	0.321		
	taud		2.516	2.533	2.575	2.577	2.570	2.592	2.593	2.556	2.490	2.475	2.482	2.533		
	Ebn at Noon		318	312	305	288	272	266	273	288	299	310	317	323		
	Edh at Noon		35	33	29	26	22	20	21	26	31	35	36	35		
(44) All-Sky Solar Radiation	RadAvg		2321	2025	1606	1112	743	581	644	910	1280	1701	2008	2264		
	RadStd		143	105	93	70	40	45	43	61	87	143	129	129		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.88	N/A	N/A	+1.57	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)			+0.90	N/A	N/A	+1.55	N/A	N/A	-29	-227	+311

Nomenclature: See separate page